

ZAHEER SK 23MID0195 HandsOnSession-30.07.26

```
In [1]: !pip install nltk
```

```
Requirement already satisfied: nltk in c:\users\skzah\anaconda3\lib\site-packages (3.9.1)  
Requirement already satisfied: click in c:\users\skzah\anaconda3\lib\site-packages (from nltk) (8.1.8)  
Requirement already satisfied: joblib in c:\users\skzah\anaconda3\lib\site-packages (from nltk) (1.4.2)  
Requirement already satisfied: regex>=2021.8.3 in c:\users\skzah\anaconda3\lib\site-packages (from nltk) (2024.11.6)  
Requirement already satisfied: tqdm in c:\users\skzah\anaconda3\lib\site-packages (from nltk) (4.67.1)  
Requirement already satisfied: colorama in c:\users\skzah\anaconda3\lib\site-packages (from click->nltk) (0.4.6)
```

```
In [3]: import nltk  
        nltk.download('brown')
```

```
[nltk_data] Downloading package brown to  
[nltk_data] C:\Users\skzah\AppData\Roaming\nltk_data...  
[nltk_data] Unzipping corpora\brown.zip.
```

```
Out[3]: True
```

```
In [5]: from nltk.corpus import brown
```

```
In [6]: brown.categories()
```

```
Out[6]: ['adventure',  
         'belles_lettres',  
         'editorial',  
         'fiction',  
         'government',  
         'hobbies',  
         'humor',  
         'learned',  
         'lore',  
         'mystery',  
         'news',  
         'religion',  
         'reviews',  
         'romance',  
         'science_fiction']
```

```
In [8]: import nltk
        nltk.download('inaugural')
        from nltk.corpus import inaugural
```

```
[nltk_data] Downloading package inaugural to
[nltk_data]   C:\Users\skzah\AppData\Roaming\nltk_data...
[nltk_data]   Package inaugural is already up-to-date!
```

```
In [9]: inaugural.fileids()
```

```
Out[9]: ['1789-Washington.txt',  
        '1793-Washington.txt',  
        '1797-Adams.txt',  
        '1801-Jefferson.txt',  
        '1805-Jefferson.txt',  
        '1809-Madison.txt',  
        '1813-Madison.txt',  
        '1817-Monroe.txt',  
        '1821-Monroe.txt',  
        '1825-Adams.txt',  
        '1829-Jackson.txt',  
        '1833-Jackson.txt',  
        '1837-VanBuren.txt',  
        '1841-Harrison.txt',  
        '1845-Polk.txt',  
        '1849-Taylor.txt',  
        '1853-Pierce.txt',  
        '1857-Buchanan.txt',  
        '1861-Lincoln.txt',  
        '1865-Lincoln.txt',  
        '1869-Grant.txt',  
        '1873-Grant.txt',  
        '1877-Hayes.txt',  
        '1881-Garfield.txt',  
        '1885-Cleveland.txt',  
        '1889-Harrison.txt',  
        '1893-Cleveland.txt',  
        '1897-McKinley.txt',  
        '1901-McKinley.txt',  
        '1905-Roosevelt.txt',  
        '1909-Taft.txt',  
        '1913-Wilson.txt',  
        '1917-Wilson.txt',  
        '1921-Harding.txt',  
        '1925-Coolidge.txt',  
        '1929-Hoover.txt',  
        '1933-Roosevelt.txt',  
        '1937-Roosevelt.txt',  
        '1941-Roosevelt.txt',  
        '1945-Roosevelt.txt',
```

```
'1949-Truman.txt',  
'1953-Eisenhower.txt',  
'1957-Eisenhower.txt',  
'1961-Kennedy.txt',  
'1965-Johnson.txt',  
'1969-Nixon.txt',  
'1973-Nixon.txt',  
'1977-Carter.txt',  
'1981-Reagan.txt',  
'1985-Reagan.txt',  
'1989-Bush.txt',  
'1993-Clinton.txt',  
'1997-Clinton.txt',  
'2001-Bush.txt',  
'2005-Bush.txt',  
'2009-Obama.txt',  
'2013-Obama.txt',  
'2017-Trump.txt',  
'2021-Biden.txt',  
'2025-Trump.txt']
```

```
In [10]: inaugural.words(fileids='2005-Bush.txt')[:20]
```

```
Out[10]: ['Vice',  
          'President',  
          'Cheney',  
          ',',  
          'Mr',  
          '.',  
          'Chief',  
          'Justice',  
          ',',  
          'President',  
          'Carter',  
          ',',  
          'President',  
          'Bush',  
          ',',  
          'President',  
          'Clinton',  
          ',',  
          'members',  
          'of']
```

```
In [11]: inaugural.words(fileids='2025-Trump.txt')[:20]
```

```
Out[11]: ['Thank',  
          'you',  
          '.',  
          'Thank',  
          'you',  
          'very',  
          'much',  
          ',',  
          'everybody',  
          '.',  
          'Wow',  
          '.',  
          'Thank',  
          'you',  
          'very',  
          ',',  
          'very',  
          'much',  
          '.',  
          'Vice']
```

```
In [12]: inaugural.words(fileids='2013-Obama.txt')[:20]
```

```
Out[12]: ['Thank',  
          'you',  
          '.',  
          'Thank',  
          'you',  
          'so',  
          'much',  
          '.',  
          'Vice',  
          'President',  
          'Biden',  
          ',',  
          'Mr',  
          '.',  
          'Chief',  
          'Justice',  
          ',',  
          'Members',  
          'of',  
          'the']
```

```
In [13]: inaugural.words(fileids='1861-Lincoln.txt')[:20]
```

```
Out[13]: ['Fellow',  
         '-',  
         'Citizens',  
         'of',  
         'the',  
         'United',  
         'States',  
         ':',  
         'In',  
         'compliance',  
         'with',  
         'a',  
         'custom',  
         'as',  
         'old',  
         'as',  
         'the',  
         'Government',  
         'itself',  
         ',']
```

```
In [15]: !pip install wordcloud matplotlib
```

Collecting wordcloud

Downloading wordcloud-1.9.6-cp313-cp313-win_amd64.whl.metadata (3.5 kB)

Requirement already satisfied: matplotlib in c:\users\skzah\anaconda3\lib\site-packages (3.10.0)

Requirement already satisfied: numpy>=1.19.3 in c:\users\skzah\anaconda3\lib\site-packages (from wordcloud) (2.1.3)

Requirement already satisfied: pillow in c:\users\skzah\anaconda3\lib\site-packages (from wordcloud) (11.1.0)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\skzah\anaconda3\lib\site-packages (from matplotlib) (1.3.1)

Requirement already satisfied: cycler>=0.10 in c:\users\skzah\anaconda3\lib\site-packages (from matplotlib) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\skzah\anaconda3\lib\site-packages (from matplotlib) (4.55.3)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\skzah\anaconda3\lib\site-packages (from matplotlib) (1.4.8)

Requirement already satisfied: packaging>=20.0 in c:\users\skzah\anaconda3\lib\site-packages (from matplotlib) (24.2)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\skzah\anaconda3\lib\site-packages (from matplotlib) (3.2.0)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\skzah\anaconda3\lib\site-packages (from matplotlib) (2.9.0.post 0)

Requirement already satisfied: six>=1.5 in c:\users\skzah\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

Downloading wordcloud-1.9.6-cp313-cp313-win_amd64.whl (306 kB)

Installing collected packages: wordcloud

Successfully installed wordcloud-1.9.6

```
In [17]: import nltk
nltk.download('stopwords')
from nltk.corpus import inaugural, stopwords
from wordcloud import WordCloud
import matplotlib.pyplot as plt
```

```
[nltk_data] Downloading package stopwords to
[nltk_data] C:\Users\skzah\AppData\Roaming\nltk_data...
[nltk_data] Unzipping corpora\stopwords.zip.
```

```
In [18]: # List available inaugural speech files
print(inaugural.fileids())

# Load Obama's 2009 inaugural speech
text = inaugural.raw('2009-Obama.txt')

# Stopwords
stop_words = set(stopwords.words('english'))

# Generate Word Cloud
wordcloud = WordCloud(
    width=1000,
```

```
height=500,  
background_color='white',  
stopwords=stop_words,  
colormap='viridis'  
).generate(text)  
  
# Display  
plt.figure(figsize=(12,6))  
plt.imshow(wordcloud, interpolation='bilinear')  
plt.axis('off')  
plt.title("Word Cloud - Barack Obama's 2009 Inaugural Address")  
plt.show()
```

```
['1789-Washington.txt', '1793-Washington.txt', '1797-Adams.txt', '1801-Jefferson.txt', '1805-Jefferson.txt', '1809-Madison.tx  
t', '1813-Madison.txt', '1817-Monroe.txt', '1821-Monroe.txt', '1825-Adams.txt', '1829-Jackson.txt', '1833-Jackson.txt', '1837-V  
anBuren.txt', '1841-Harrison.txt', '1845-Polk.txt', '1849-Taylor.txt', '1853-Pierce.txt', '1857-Buchanan.txt', '1861-Lincoln.tx  
t', '1865-Lincoln.txt', '1869-Grant.txt', '1873-Grant.txt', '1877-Hayes.txt', '1881-Garfield.txt', '1885-Cleveland.txt', '1889-  
Harrison.txt', '1893-Cleveland.txt', '1897-McKinley.txt', '1901-McKinley.txt', '1905-Roosevelt.txt', '1909-Taft.txt', '1913-Wil  
son.txt', '1917-Wilson.txt', '1921-Harding.txt', '1925-Coolidge.txt', '1929-Hoover.txt', '1933-Roosevelt.txt', '1937-Roosevelt.  
txt', '1941-Roosevelt.txt', '1945-Roosevelt.txt', '1949-Truman.txt', '1953-Eisenhower.txt', '1957-Eisenhower.txt', '1961-Kenned  
y.txt', '1965-Johnson.txt', '1969-Nixon.txt', '1973-Nixon.txt', '1977-Carter.txt', '1981-Reagan.txt', '1985-Reagan.txt', '1989-  
Bush.txt', '1993-Clinton.txt', '1997-Clinton.txt', '2001-Bush.txt', '2005-Bush.txt', '2009-Obama.txt', '2013-Obama.txt', '2017-  
Trump.txt', '2021-Biden.txt', '2025-Trump.txt']
```

[illegible]

```
Out[19]: FreqDist({'and': 10, 'intelligence': 5, 'technology': 5, 'education': 5, 'students': 5, 'Artificial': 4, 'to': 4, 'improves': 4, 'artificial': 4, 'data': 4, ...})
```

```
In [20]: import nltk
```

```
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt_tab to  
[nltk_data] C:\Users\skzah\AppData\Roaming\nltk_data...  
[nltk_data] Unzipping tokenizers\punkt_tab.zip.
```

Out[20]: True

```
In [21]: import nltk  
from nltk.tokenize import word_tokenize  
from nltk.probability import FreqDist  
from nltk.corpus import stopwords  
import string
```

```
In [22]: text = """  
Artificial intelligence is transforming the world. Artificial intelligence helps people solve problems,  
improve healthcare, enhance education, and develop innovative technology. Technology continues to grow  
because technology supports communication, technology improves productivity, and technology creates  
opportunities. Education plays an important role because education develops knowledge, education improves  
skills, and education prepares students for the future. Data is essential in artificial intelligence because  
data enables learning, data improves predictions, and data supports decision making. Innovation drives  
technology, innovation creates solutions, and innovation encourages progress. Students learn technology,  
students analyze data, students develop skills, and students contribute to innovation. Artificial intelligence,  
technology, education, data, innovation, and students together create a smarter future. Technology,  
technology, technology and artificial intelligence continue to shape society. Education, education,  
education empowers every student. Data, data, data improves every intelligent system. Innovation,  
innovation, innovation drives economic growth and scientific progress. Artificial intelligence,  
artificial intelligence, artificial intelligence will continue to transform technology and improve education  
for future generations.  
"""  
  
# Convert to Lowercase  
text = text.lower()  
  
# Tokenize  
words = word_tokenize(text)  
  
# Remove punctuation and stopwords  
stop_words = set(stopwords.words('english'))  
  
filtered_words = [
```

```
    word for word in words
    if word not in stop_words and word not in string.punctuation
]

# Frequency Distribution
fdist = FreqDist(filtered_words)

# Display all frequencies
print("Word Frequency Distribution:\n")
for word, freq in fdist.most_common():
    print(f"{word:<15} : {freq}")

# Top 10 most frequent words
print("\nTop 10 Most Frequent Words:\n")
for word, freq in fdist.most_common(10):
    print(f"{word:<15} : {freq}")

# Plot frequency distribution
fdist.plot(10, title="Top 10 Most Frequent Words")
```

Word Frequency Distribution:

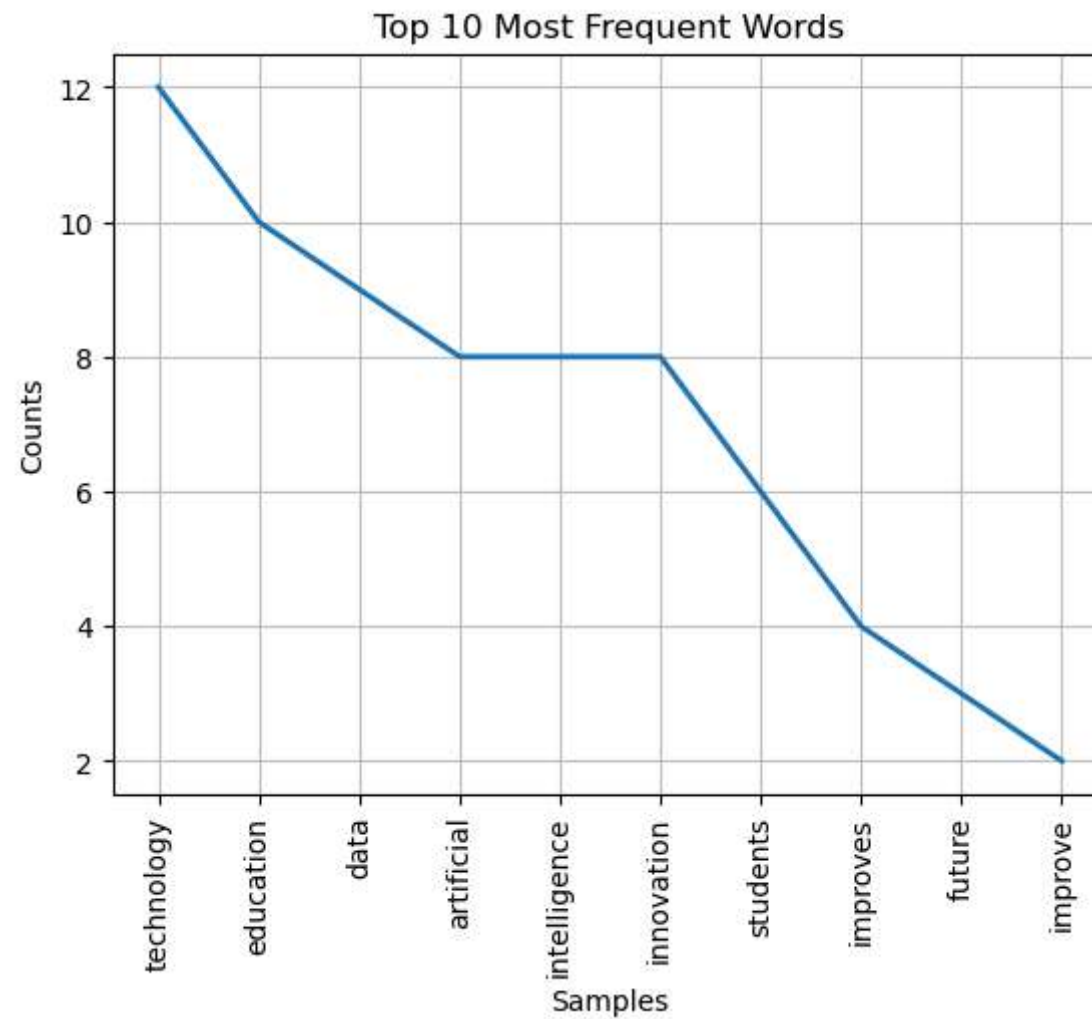
technology	: 12
education	: 10
data	: 9
artificial	: 8
intelligence	: 8
innovation	: 8
students	: 6
improves	: 4
future	: 3
improve	: 2
develop	: 2
supports	: 2
creates	: 2
skills	: 2
drives	: 2
progress	: 2
continue	: 2
every	: 2
transforming	: 1
world	: 1
helps	: 1
people	: 1
solve	: 1
problems	: 1
healthcare	: 1
enhance	: 1
innovative	: 1
continues	: 1
grow	: 1
communication	: 1
productivity	: 1
opportunities	: 1
plays	: 1
important	: 1
role	: 1
develops	: 1
knowledge	: 1
prepares	: 1
essential	: 1

enables	: 1
learning	: 1
predictions	: 1
decision	: 1
making	: 1
solutions	: 1
encourages	: 1
learn	: 1
analyze	: 1
contribute	: 1
together	: 1
create	: 1
smarter	: 1
shape	: 1
society	: 1
empowers	: 1
student	: 1
intelligent	: 1
system	: 1
economic	: 1
growth	: 1
scientific	: 1
transform	: 1
generations	: 1

Top 10 Most Frequent Words:

technology	: 12
education	: 10
data	: 9
artificial	: 8
intelligence	: 8
innovation	: 8
students	: 6
improves	: 4
future	: 3
improve	: 2

Out[22]: <Axes: title={'center': 'Top 10 Most Frequent Words'}, xlabel='Samples', ylabel='Counts'>



```
In [23]: import nltk
from nltk.corpus import inaugural, stopwords
from nltk.tokenize import word_tokenize
from nltk.probability import FreqDist
import string
```

```
In [24]: # Load Washington's inaugural speech
text = inaugural.raw('1789-Washington.txt')
```

```
# Convert to Lowercase
text = text.lower()

# Tokenize
words = word_tokenize(text)

# Remove stopwords and punctuation
stop_words = set(stopwords.words('english'))
words = [
    word for word in words
    if word not in stop_words and word not in string.punctuation
]

# Frequency Distribution
fdist = FreqDist(words)

# Top 20 frequent words
print("Top 20 Most Frequent Words:\n")
for word, freq in fdist.most_common(20):
    print(f"{word:<15} {freq}")
```

Top 20 Most Frequent Words:

every	9
government	8
public	6
may	6
present	5
country	5
one	4
citizens	4
ought	4
duty	4
people	4
united	4
since	4
could	3
hand	3
nature	3
happiness	3
great	3
good	3
less	3

```
In [26]: from nltk.probability import ConditionalFreqDist
```

```
        cfd = ConditionalFreqDist(  
            (len(word), word) for word in text.split()  
        )  
  
        cfd[6]
```

```
Out[26]: FreqDist({'public': 6, 'united': 4, 'nature': 3, 'people': 3, 'myself': 3, 'former': 2, 'fellow': 2, 'assure': 2, 'nation':  
2, 'humble': 2, ...})
```

```
In [27]: import nltk  
        from nltk.corpus import inaugural  
        from nltk.tokenize import word_tokenize  
        from nltk.corpus import stopwords  
        import string  
        from collections import Counter
```

```
In [28]: stop_words = set(stopwords.words('english'))

results = []

for fileid in inaugural.fileids():

    # President's name
    president = fileid.split("-")[1].replace(".txt", "")

    # Read speech
    text = inaugural.raw(fileid).lower()

    # Tokenize
    words = word_tokenize(text)

    # Keep only alphabetic words
    words = [
        w for w in words
        if w.isalpha() and w not in stop_words
    ]

    total_words = len(words)

    # High-sounding words (length >= 8)
    long_words = [w for w in words if len(w) >= 8]

    count = len(long_words)
    percentage = (count / total_words) * 100

    results.append((president, fileid[:4], count, percentage))

# Sort by percentage of long words
results = sorted(results, key=lambda x: x[3], reverse=True)

print(f"{'President':15} {'Year':6} {'Long Words':12} {'Percentage'}")
print("-"*50)

for r in results:
    print(f"{r[0]:15} {r[1]:6} {r[2]:12} {r[3]:.2f}%")
```

President	Year	Long Words	Percentage

Washington	1793	28	46.67%
Taylor	1849	241	46.62%
Jackson	1829	240	45.71%
Washington	1789	291	45.05%
Adams	1825	610	44.99%
Madison	1809	231	44.59%
Adams	1797	467	44.06%
VanBuren	1837	826	43.98%
Cleveland	1893	422	43.19%
Hoover	1929	724	42.99%
Cleveland	1885	337	42.23%
Taft	1909	1068	41.83%
Pierce	1853	653	41.83%
Harrison	1841	1570	41.63%
Jackson	1833	226	41.62%
Hayes	1877	488	41.29%
McKinley	1901	429	40.97%
Polk	1845	922	40.89%
Madison	1813	221	40.62%
Harrison	1889	819	39.51%
Monroe	1817	605	39.06%
McKinley	1897	742	39.01%
Jefferson	1805	390	38.84%
Monroe	1821	771	38.30%
Harding	1921	647	38.24%
Garfield	1881	534	37.98%
Coolidge	1925	709	37.96%
Lincoln	1861	608	37.79%
Buchanan	1857	505	37.49%
Grant	1873	216	35.53%
Jefferson	1801	282	34.73%
Grant	1869	183	34.53%
Truman	1949	380	34.02%
Roosevelt	1933	309	33.81%
Roosevelt	1937	292	32.81%
Wilson	1913	227	28.55%
Reagan	1981	318	28.24%
Carter	1977	160	28.07%
Trump	2017	198	27.97%

Roosevelt	1905	122	27.85%
Trump	2025	384	27.83%
Clinton	1993	217	27.64%
Roosevelt	1941	166	26.90%
Eisenhower	1953	317	26.20%
Wilson	1917	170	25.64%
Obama	2013	251	25.15%
Lincoln	1865	85	25.15%
Reagan	1985	320	25.12%
Bush	2001	194	25.10%
Bush	2005	246	24.87%
Clinton	1997	271	24.52%
Nixon	1973	193	23.48%
Obama	2009	249	21.80%
Nixon	1969	218	21.73%
Johnson	1965	145	21.20%
Kennedy	1961	144	21.05%
Biden	2021	243	20.91%
Roosevelt	1945	51	20.65%
Bush	1989	213	19.87%
Eisenhower	1957	150	18.70%

```
In [29]: from collections import Counter

for fileid in inaugural.fileids():

    president = fileid.split("-")[1].replace(".txt", "")

    text = inaugural.raw(fileid).lower()

    words = word_tokenize(text)

    words = [
        w for w in words
        if w.isalpha()
        and w not in stopwords.words("english")
        and len(w) >= 8
    ]

    common = Counter(words).most_common(10)
```

```
print(f"\n{fileid}")  
print(common)
```

1789-Washington.txt

[('government', 8), ('citizens', 4), ('happiness', 3), ('measures', 3), ('representatives', 2), ('immutable', 2), ('rendered', 2), ('necessary', 2), ('qualifications', 2), ('administration', 2)]

1793-Washington.txt

[('citizens', 1), ('functions', 1), ('magistrate', 1), ('occasion', 1), ('endeavor', 1), ('entertain', 1), ('distinguished', 1), ('confidence', 1), ('previous', 1), ('execution', 1)]

1797-Adams.txt

[('government', 16), ('constitution', 8), ('congress', 6), ('national', 4), ('pleasing', 4), ('knowledge', 4), ('virtuous', 4), ('interest', 4), ('legislature', 3), ('certainly', 3)]

1801-Jefferson.txt

[('government', 12), ('principle', 6), ('citizens', 5), ('happiness', 4), ('confidence', 4), ('industry', 3), ('commerce', 3), ('political', 3), ('republican', 3), ('principles', 3)]

1805-Jefferson.txt

[('citizens', 10), ('constitution', 6), ('interests', 4), ('interest', 4), ('therefore', 4), ('measures', 4), ('occasion', 3), ('principles', 3), ('expenses', 3), ('collected', 3)]

1809-Madison.txt

[('confidence', 3), ('improvements', 3), ('examples', 2), ('circumstances', 2), ('gratitude', 2), ('difficulties', 2), ('national', 2), ('rendered', 2), ('resources', 2), ('agriculture', 2)]

1813-Madison.txt

[('political', 3), ('citizens', 3), ('government', 3), ('honorable', 3), ('obligations', 2), ('distinguished', 2), ('confidence', 2), ('consideration', 2), ('reliance', 2), ('received', 2)]

1817-Monroe.txt

[('government', 21), ('principles', 9), ('national', 8), ('importance', 6), ('faithful', 6), ('interests', 5), ('interest', 5), ('preserve', 5), ('connected', 4), ('executive', 4)]

1821-Monroe.txt

[('citizens', 14), ('government', 12), ('commerce', 11), ('necessary', 7), ('sovereignty', 6), ('purposes', 6), ('measures', 6), ('especially', 6), ('permanent', 5), ('respecting', 5)]

1825-Adams.txt

[('government', 17), ('constitution', 8), ('national', 6), ('political', 6), ('interests', 6), ('performance', 4), ('purposes', 4), ('blessings', 4), ('condition', 4), ('generation', 4)]

1829-Jackson.txt

[('government', 6), ('national', 4), ('interests', 3), ('generally', 3), ('citizens', 2), ('accountability', 2), ('constitution', 2), ('endeavor', 2), ('congress', 2), ('executive', 2)]

1833-Jackson.txt

[('government', 13), ('preservation', 4), ('exercise', 4), ('citizens', 3), ('confidence', 3), ('importance', 3), ('continue', 3), ('american', 2), ('expressed', 2), ('expression', 2)]

1837-VanBuren.txt

[('institutions', 16), ('government', 15), ('experience', 8), ('principles', 7), ('confidence', 7), ('citizens', 6), ('countrymen', 6), ('political', 5), ('prosperity', 5), ('opinions', 5)]

1841-Harrison.txt

[('constitution', 36), ('government', 36), ('executive', 25), ('character', 17), ('citizens', 16), ('principle', 11), ('president', 11), ('congress', 11), ('principles', 10), ('institutions', 10)]

1845-Polk.txt

[('government', 45), ('constitution', 15), ('interests', 14), ('protection', 12), ('citizens', 10), ('institutions', 9), ('extended', 8), ('independent', 7), ('millions', 7), ('exercise', 6)]

1849-Taylor.txt

[('government', 7), ('constitution', 4), ('interests', 4), ('congress', 4), ('measures', 3), ('american', 2), ('confidence', 2), ('republic', 2), ('discharge', 2), ('position', 2)]

1853-Pierce.txt

[('government', 10), ('administration', 6), ('constitutional', 6), ('national', 6), ('position', 5), ('republic', 5), ('constitution', 5), ('confidence', 4), ('obligations', 4), ('strength', 4)]

1857-Buchanan.txt

[('constitution', 17), ('government', 13), ('question', 11), ('congress', 9), ('territory', 8), ('domestic', 6), ('necessary', 6), ('citizens', 5), ('political', 5), ('importance', 5)]

1861-Lincoln.txt

[('constitution', 24), ('government', 18), ('national', 9), ('constitutional', 8), ('authority', 7), ('administration', 6), ('provision', 6), ('minority', 6), ('congress', 5), ('surrendered', 5)]

1865-Lincoln.txt

[('interest', 3), ('offenses', 3), ('occasion', 2), ('altogether', 2), ('conflict', 2), ('answered', 2), ('appearing', 1), ('presidential', 1), ('extended', 1), ('statement', 1)]

1869-Grant.txt

[('determination', 3), ('greatest', 3), ('interest', 3), ('national', 3), ('government', 3), ('necessary', 3), ('question', 3),

('citizens', 2), ('constitution', 2), ('questions', 2)]

1873-Grant.txt

[('executive', 4), ('commerce', 4), ('government', 4), ('territory', 4), ('proposition', 4), ('extension', 3), ('throughout', 3), ('influence', 3), ('question', 3), ('relations', 2)]

1877-Hayes.txt

[('government', 15), ('political', 10), ('citizens', 7), ('interests', 7), ('principles', 6), ('constitution', 6), ('questions', 5), ('condition', 5), ('question', 5), ('administration', 4)]

1881-Garfield.txt

[('government', 20), ('constitution', 15), ('authority', 9), ('congress', 9), ('national', 7), ('administration', 7), ('suffrage', 6), ('blessings', 5), ('protection', 5), ('republic', 4)]

1885-Cleveland.txt

[('government', 16), ('constitution', 8), ('interests', 7), ('citizens', 5), ('partisan', 5), ('national', 4), ('prosperity', 4), ('american', 4), ('application', 3), ('benefits', 3)]

1889-Harrison.txt

[('constitution', 11), ('government', 10), ('congress', 8), ('interest', 7), ('friendly', 7), ('condition', 6), ('american', 6), ('european', 6), ('necessary', 6), ('executive', 5)]

1893-Cleveland.txt

[('government', 14), ('american', 8), ('national', 6), ('countrymen', 5), ('interests', 5), ('citizens', 4), ('confidence', 4), ('institutions', 4), ('political', 4), ('strength', 3)]

1897-McKinley.txt

[('government', 23), ('congress', 18), ('american', 9), ('legislation', 9), ('business', 8), ('citizens', 7), ('president', 6), ('constant', 6), ('interests', 6), ('question', 5)]

1901-McKinley.txt

[('government', 14), ('congress', 9), ('executive', 9), ('american', 7), ('national', 5), ('inhabitants', 5), ('obligations', 4), ('faithful', 4), ('republic', 4), ('relations', 3)]

1905-Roosevelt.txt

[('problems', 5), ('republic', 4), ('conditions', 3), ('responsibility', 3), ('relations', 3), ('children', 3), ('foundations', 2), ('national', 2), ('continent', 2), ('existence', 2)]

1909-Taft.txt

[('government', 26), ('business', 22), ('congress', 14), ('interstate', 10), ('legislation', 10), ('necessary', 10), ('executive', 9), ('administration', 8), ('commerce', 8), ('consideration', 8)]

1913-Wilson.txt

[('government', 9), ('democratic', 3), ('question', 3), ('familiar', 3), ('enterprise', 3), ('industrial', 3), ('children', 3), ('interpret', 2), ('comprehend', 2), ('individual', 2)]

1917-Wilson.txt

[('national', 5), ('principles', 5), ('interest', 4), ('citizens', 3), ('industrial', 3), ('politics', 3), ('influence', 3), ('immediate', 3), ('together', 3), ('opportunity', 3)]

1921-Harding.txt

[('civilization', 12), ('government', 11), ('republic', 10), ('american', 8), ('understanding', 7), ('national', 6), ('opportunity', 6), ('responsibility', 5), ('progress', 5), ('economic', 5)]

1925-Coolidge.txt

[('government', 14), ('independence', 7), ('property', 7), ('conditions', 6), ('resources', 6), ('continue', 6), ('necessary', 6), ('political', 6), ('determined', 5), ('humanity', 5)]

1929-Hoover.txt

[('government', 24), ('progress', 16), ('enforcement', 10), ('citizens', 9), ('education', 7), ('american', 6), ('responsibility', 6), ('business', 6), ('cooperation', 6), ('economic', 6)]

1933-Roosevelt.txt

[('national', 9), ('leadership', 7), ('emergency', 4), ('congress', 4), ('measures', 4), ('respects', 4), ('discipline', 4), ('essential', 3), ('government', 3), ('important', 3)]

1937-Roosevelt.txt

[('government', 15), ('democracy', 8), ('progress', 7), ('millions', 6), ('economic', 5), ('republic', 4), ('problems', 4), ('individual', 3), ('practical', 3), ('democratic', 3)]

1941-Roosevelt.txt

[('democracy', 9), ('government', 4), ('something', 4), ('continent', 3), ('national', 2), ('washington', 2), ('together', 2), ('disruption', 2), ('measured', 2), ('americans', 2)]

1945-Roosevelt.txt

[('democracy', 2), ('presence', 2), ('mistakes', 2), ('remember', 2), ('president', 1), ('understand', 1), ('inauguration', 1), ('americans', 1), ('together', 1), ('essential', 1)]

1949-Truman.txt

[('security', 9), ('democracy', 9), ('countries', 8), ('communism', 6), ('economic', 6), ('resources', 6), ('recovery', 5), ('international', 5), ('strengthen', 5), ('american', 4)]

1953-Eisenhower.txt

[('strength', 10), ('everywhere', 4), ('political', 4), ('productivity', 4), ('principles', 4), ('economic', 4), ('security', 4), ('dedication', 3), ('citizens', 3), ('responsibility', 3)]

1957-Eisenhower.txt

[('strength', 3), ('equality', 3), ('progress', 3), ('wherever', 2), ('principles', 2), ('purposes', 2), ('blessings', 2), ('industry', 2), ('knowledge', 2), ('material', 2)]

1961-Kennedy.txt

[('citizens', 5), ('president', 4), ('americans', 4), ('generation', 3), ('forebears', 2), ('revolution', 2), ('committed', 2), ('powerful', 2), ('supporting', 2), ('alliance', 2)]

1965-Johnson.txt

[('american', 6), ('covenant', 6), ('generation', 4), ('together', 3), ('children', 2), ('different', 2), ('stranger', 2), ('neighbors', 2), ('resources', 2), ('judgment', 2)]

1969-Nixon.txt

[('together', 8), ('government', 5), ('celebrate', 4), ('president', 3), ('american', 3), ('rhetoric', 3), ('brothers', 3), ('beginning', 2), ('centuries', 2), ('possible', 2)]

1973-Nixon.txt

[('responsibility', 11), ('government', 10), ('together', 7), ('policies', 7), ('responsibilities', 5), ('progress', 5), ('structure', 3), ('challenges', 3), ('individual', 3), ('conflict', 2)]

1977-Carter.txt

[('strength', 7), ('together', 7), ('government', 4), ('president', 3), ('american', 3), ('physical', 2), ('ceremony', 2), ('principles', 2), ('inauguration', 2), ('beginning', 2)]

1981-Reagan.txt

[('government', 16), ('americans', 9), ('president', 5), ('national', 5), ('children', 5), ('strength', 5), ('economic', 4), ('citizens', 3), ('maintaining', 3), ('political', 3)]

1985-Reagan.txt

[('government', 16), ('americans', 7), ('together', 7), ('american', 7), ('citizens', 6), ('national', 6), ('progress', 6), ('economic', 4), ('increase', 4), ('security', 4)]

1989-Bush.txt

[('president', 6), ('government', 5), ('democracy', 5), ('children', 4), ('washington', 3), ('expression', 3), ('remember', 3), ('strength', 3), ('engagement', 3), ('generation', 3)]

1993-Clinton.txt

```
[('americans', 9), ('american', 5), ('democracy', 4), ('generation', 4), ('millions', 4), ('government', 4), ('celebrate', 3), ('sacrifice', 3), ('powerful', 3), ('children', 3)]
```

1997-Clinton.txt

```
[('government', 10), ('american', 9), ('citizens', 7), ('americans', 7), ('children', 6), ('together', 6), ('democracy', 4), ('responsibility', 4), ('greatest', 4), ('millions', 3)]
```

2001-Bush.txt

```
[('citizens', 9), ('americans', 5), ('american', 4), ('sometimes', 4), ('civility', 4), ('character', 4), ('children', 4), ('government', 4), ('president', 3), ('everyone', 3)]
```

2005-Bush.txt

```
[('americans', 8), ('citizens', 6), ('president', 4), ('determined', 3), ('defended', 3), ('security', 3), ('institutions', 3), ('governments', 3), ('permanent', 3), ('character', 3)]
```

2009-Obama.txt

```
[('generation', 5), ('americans', 3), ('prosperity', 3), ('question', 3), ('government', 3), ('generations', 3), ('children', 3), ('cooperation', 2), ('throughout', 2), ('founding', 2)]
```

2013-Obama.txt

```
[('together', 7), ('citizens', 6), ('generation', 6), ('requires', 5), ('american', 5), ('complete', 5), ('happiness', 4), ('government', 4), ('americans', 4), ('enduring', 3)]
```

2017-Trump.txt

```
[('american', 11), ('president', 5), ('protected', 5), ('americans', 4), ('citizens', 4), ('together', 4), ('everyone', 4), ('government', 3), ('factories', 3), ('families', 3)]
```

2021-Biden.txt

```
[('democracy', 11), ('americans', 10), ('american', 8), ('president', 7), ('children', 6), ('together', 5), ('centuries', 4), ('strength', 4), ('understand', 4), ('challenge', 3)]
```

2025-Trump.txt

```
[('american', 13), ('government', 11), ('president', 9), ('citizens', 8), ('americans', 7), ('immediately', 5), ('together', 5), ('administration', 4), ('national', 4), ('greatest', 4)]
```

take a speech and find cfd

```
In [33]: import nltk

nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to  
[nltk_data] C:\Users\skzah\AppData\Roaming\nltk_data...  
[nltk_data] Unzipping taggers\averaged_perceptron_tagger_eng.zip.
```

Out[33]: True

```
In [35]: import nltk  
  
# If text is your complete speech stored as a single string  
text = text.lower()  
  
sentences = nltk.sent_tokenize(text)  
  
print(sentences)
```

['thank you.', 'thank you very much, everybody.', 'wow.', 'thank you very, very much.', 'vice president vance, speaker johnson, senator thune, chief justice roberts, justices of the supreme court of the united states, president clinton, president bush, president obama, president biden, vice president harris, and my fellow citizens, the golden age of america begins right now.', 'from this day forward, our country will flourish and be respected again all over the world.', 'we will be the envy of every nation, and we will not allow ourselves to be taken advantage of any longer.', 'during every single day of the trump administration, i will, very simply, put america first.', 'our sovereignty will be reclaimed.', 'our safety will be restored.', 'the scales of justice will be rebalanced.', 'the vicious, violent, and unfair weaponization of the justice department and our government will end.', 'and our top priority will be to create a nation that is proud, prosperous, and free.', 'america will soon be greater, stronger, and far more exceptional than ever before.', 'i return to the presidency confident and optimistic that we are at the start of a thrilling new era of national success.', 'a tide of change is sweeping the country, sunlight is pouring over the entire world, and america has the chance to seize this opportunity like never before.', 'but first, we must be honest about the challenges we face.', 'while they are plentiful, they will be annihilated by this great momentum that the world is now witnessing in the united states of america.', 'as we gather today, our government confronts a crisis of trust.', 'for many years, a radical and corrupt establishment has extracted power and wealth from our citizens while the pillars of our society lay broken and seemingly in complete disrepair.', 'we now have a government that cannot manage even a simple crisis at home while, at the same time, stumbling into a continuing catalogue of catastrophic events abroad.', 'it fails to protect our magnificent, law-abiding american citizens but provides sanctuary and protection for dangerous criminals, many from prisons and mental institutions, that have illegally entered our country from all over the world.', 'we have a government that has given unlimited funding to the defense of foreign borders but refuses to defend american borders or, more importantly, its own people.', 'our country can no longer deliver basic services in times of emergency, as recently shown by the wonderful people of north carolina who have been treated so badly and other states who are still suffering from a hurricane that took place many months ago or, more recently, los angeles, where we are watching fires still tragically burn from weeks ago without even a token of defense.', 'they're raging through the houses and communities, even affecting some of the wealthiest and most powerful individuals in our country some of whom are sitting here right now.', 'they don't have a home any longer.', 'that's interesting.', 'but we can't let this happen.', 'everyone is unable to do anything about it.', 'that's going to change.', 'we have a public health system that does not deliver in times of disaster, yet more money is spent on it than any country anywhere in the world.', 'and we have an education system that teaches our children to be ashamed of themselves in many cases, to hate our country despite the love that we try so desperately to provide to them.', 'all of this will change starting today, and it will change very quickly.', 'my recent election is a mandate to completely and totally reverse a horrible betrayal and all of these many betrayals that have taken place and to give the people back their faith, their wealth, their democracy, and, indeed, their freedom.', 'from this moment on, america's decline is over.', 'our liberties and our nation's glorious destiny will no longer be denied.', 'and we will immediately restore the integrity, competency, and loyalty of america's government.', 'over the past eight years, i have been tested and challenged more than any president in our 250-year history, and i've learned a lot along the way.', 'the journey to reclaim our republic has not been an easy one that, i can tell you.', 'those who wish to stop our cause have tried to take my freedom and, indeed, to take my life.', 'just a few months ago, in a beautiful pennsylvania field, an assassin's bullet ripped through my ear.', 'but i felt then and believe even more so now that my life was saved for a reason.', 'i was saved by god to make america great again.', 'thank you.', 'thank you.', 'thank you very much.', 'that is why each day under our administration of american patriots, we will be working to meet every crisis with dignity and power and strength.', 'we will move with purpose and speed to bring back hope, prosperity, safety, and peace for citizens of every race, religion, color, and creed.', 'for american citizens, january 20th, 2025, is liberation day.', 'it is my hope that our recent presidential election will be remembered as the greatest and most

consequential election in the history of our country.', 'as our victory showed, the entire nation is rapidly unifying behind our agenda with dramatic increases in support from virtually every element of our society: young and old, men and women, african americans, hispanic americans, asian americans, urban, suburban, rural.', 'and very importantly, we had a powerful win in all seven swing states and the popular vote, we won by millions of people.', 'to the black and hispanic communities, i want to thank you for the tremendous outpouring of love and trust that you have shown me with your vote.', 'we set records, and i will not forget it.', 'iâ\x80\x99ve heard your voices in the campaign, and i look forward to working with you in the years to come.', 'today is martin luther king day.', 'and his honorâ\x80\x94this will be a great honor.', 'but in his honor, we will strive together to make his dream a reality.', 'we will make his dream come true.', 'thank you.', 'thank you.', 'thank you.', 'national unity is now returning to america, and confidence and pride is soaring like never before.', 'in everything we do, my administration will be inspired by a strong pursuit of excellence and unrelenting success.', 'we will not forget our country, we will not forget our constitution, and we will not forget our god.', 'canâ\x80\x99t do that.', 'today, i will sign a series of historic executive orders.', 'with these actions, we will begin the complete restoration of america and the revolution of common sense.', 'itâ\x80\x99s all about common sense.', 'first, i will declare a national emergency at our southern border.', 'all illegal entry will immediately be halted, and we will begin the process of returning millions and millions of criminal aliens back to the places from which they came.', 'we will reinstate my remain in mexico policy.', 'i will end the practice of catch and release.', 'and i will send troops to the southern border to repel the disastrous invasion of our country.', 'under the orders i sign today, we will also be designating the cartels as foreign terrorist organizations.', 'and by invoking the alien enemies act of 1798, i will direct our government to use the full and immense power of federal and state law enforcement to eliminate the presence of all foreign gangs and criminal networks bringing devastating crime to us soil, including our cities and inner cities.', 'as commander in chief, i have no higher responsibility than to defend our country from threats and invasions, and that is exactly what i am going to do.', 'we will do it at a level that nobody has ever seen before.', 'next, i will direct all members of my cabinet to marshal the vast powers at their disposal to defeat what was record inflation and rapidly bring down costs and prices.', 'the inflation crisis was caused by massive overspending and escalating energy prices, and that is why today i will also declare a national energy emergency.', 'we will drill, baby, drill.', 'america will be a manufacturing nation once again, and we have something that no other manufacturing nation will ever haveâ\x80\x94the largest amount of oil and gas of any country on earthâ\x80\x94and we are going to use it.', 'weâ\x80\x99ll use it.', 'we will bring prices down, fill our strategic reserves up again right to the top, and export american energy all over the world.', 'we will be a rich nation again, and it is that liquid gold under our feet that will help to do it.', 'with my actions today, we will end the green new deal, and we will revoke the electric vehicle mandate, saving our auto industry and keeping my sacred pledge to our great american autoworkers.', 'in other words, youâ\x80\x99ll be able to buy the car of your choice.', 'we will build automobiles in america again at a rate that nobody could have dreamt possible just a few years ago.', 'and thank you to the autoworkers of our nation for your inspiring vote of confidence.', 'we did tremendously with their vote.', 'i will immediately begin the overhaul of our trade system to protect american workers and families.', 'instead of taxing our citizens to enrich other countries, we will tariff and tax foreign countries to enrich our citizens.', 'for this purpose, we are establishing the external revenue service to collect all tariffs, duties, and revenues.', 'it will be massive amounts of money pouring into our treasury, coming from foreign sources.', 'the american dream will soon be back and thriving like never before.', 'to restore competence and effectiveness to our federal government, my administration will establish the brand-new department of government efficiency.', 'after years and years of illegal and unconstitutional federal efforts to restrict free expression, i also will sign an executive order to immediately stop all government censorship and bring back free speech to america.', 'never again will the immense power of the state be weaponized to persecute political opponentsâ\x80\x94something i know something about.', 'we will not allow that to happen.', 'it will not happen again.', 'under my leadership, we will restore fair, equal, and impartial justice under the constitutional rule of law.', 'and we are going

g to bring law and order back to our cities.', 'this week, i will also end the government policy of trying to socially engineer race and gender into every aspect of public and private life.', 'we will forge a society that is colorblind and merit-based.', 'as of today, it will henceforth be the official policy of the united states government that there are only two genders: male and female.', 'this week, i will reinstate any service members who were unjustly expelled from our military for objecting to the covid vaccine mandate with full back pay.', 'and i will sign an order to stop our warriors from being subjected to radical political theories and social experiments while on duty.', 'itâ\x80\x99s going to end immediately.', 'our armed forces will be freed to focus on their sole mission: defeating americaâ\x80\x99s enemies.', 'like in 2017, we will again build the strongest military the world has ever seen.', 'we will measure our success not only by the battles we win but also by the wars that we endâ\x80\x94and perhaps most importantly, the wars we never get into.', 'my proudest legacy will be that of a peacemaker and unifier.', 'thatâ\x80\x99s what i want to be: a peacemaker and a unifier.', 'iâ\x80\x99m pleased to say that as of yesterday, one day before i assumed office, the hostages in the middle east are coming back home to their families.', 'thank you.', 'america will reclaim its rightful place as the greatest, most powerful, most respected nation on earth, inspiring the awe and admiration of the entire world.', 'a short time from now, we are going to be changing the name of the gulf of mexico to the gulf of america, and we will restore the name of a great president, william mckinley, to mount mckinley, where it should be and where it belongs.', 'president mckinley made our country very rich through tariffs and through talentâ\x80\x94he was a natural businessmanâ\x80\x94and gave teddy roosevelt the money for many of the great things he did, including the panama canal, which has foolishly been given to the country of panama after the united statesâ\x80\x94the united states, i mean, think of thisâ\x80\x94spent more money than ever spent on a project before and lost 38,000 lives in the building of the panama canal.', 'we have been treated very badly from this foolish gift that should have never been made, and panamaâ\x80\x99s promise to us has been broken.', 'the purpose of our deal and the spirit of our treaty has been totally violated.', 'american ships are being severely overcharged and not treated fairly in any way, shape, or form.', 'and that includes the united states navy.', 'and above all, china is operating the panama canal.', 'and we didnâ\x80\x99t give it to china.', 'we gave it to panama, and weâ\x80\x99re taking it back.', 'above all, my message to americans today is that it is time for us to once again act with courage, vigor, and the vitality of historyâ\x80\x99s greatest civilization.', 'so, as we liberate our nation, we will lead it to new heights of victory and success.', 'we will not be deterred.', 'together, we will end the chronic disease epidemic and keep our children safe, healthy, and disease-free.', 'the united states will once again consider itself a growing nationâ\x80\x94one that increases our wealth, expands our territory, builds our cities, raises our expectations, and carries our flag into new and beautiful horizons.', 'and we will pursue our manifest destiny into the stars, launching american astronauts to plant the stars and stripes on the planet mars.', 'ambition is the lifeblood of a great nation, and, right now, our nation is more ambitious than any other.', 'thereâ\x80\x99s no nation like our nation.', 'americans are explorers, builders, innovators, entrepreneurs, and pioneers.', 'the spirit of the frontier is written into our hearts.', 'the call of the next great adventure resounds from within our souls.', 'our american ancestors turned a small group of colonies on the edge of a vast continent into a mighty republic of the most extraordinary citizens on earth.', 'no one comes close.', 'americans pushed thousands of miles through a rugged land of untamed wilderness.', 'they crossed deserts, scaled mountains, braved untold dangers, won the wild west, ended slavery, rescued millions from tyranny, lifted billions from poverty, harnessed electricity, split the atom, launched mankind into the heavens, and put the universe of human knowledge into the palm of the human hand.', 'if we work together, there is nothing we cannot do and no dream we cannot achieve.', 'many people thought it was impossible for me to stage such a historic political comeback.', 'but as you see today, here i am.', 'the american people have spoken.', 'i stand before you now as proof that you should never believe that something is impossible to do.', 'in america, the impossible is what we do best.', 'from new york to los angeles, from philadelphia to phoenix, from chicago to miami, from houston to right here in washington, dc, our country was forged and built by the generations of patriots who gave everything they had for our rights and for our freedom.', 'they were farmers and soldiers, cowboys and factory

y workers, steelworkers and coal miners, police officers and pioneers who pushed onward, marched forward, and let no obstacle defeat their spirit or their pride.', 'together, they laid down the railroads, raised up the skyscrapers, built great highways, won two world wars, defeated fascism and communism, and triumphed over every single challenge that they faced.', 'after all we have been through together, we stand on the verge of the four greatest years in american history.', 'with your help, we will restore america promise and we will rebuild the nation that we loveâ\x80\x94and we love it so much.', 'we are one people, one family, and one glorious nation under god.', 'so, to every parent who dreams for their child and every child who dreams for their future, i am with you, i will fight for you, and i will win for you.', 'weâ\x80\x99re going to win like never before.', 'thank you.', 'thank you.', 'thank you.', 'thank you.', 'in recent years, our nation has suffered greatly.', 'but we are going to bring it back and make it great again, greater than ever before.', 'we will be a nation like no other, full of compassion, courage, and exceptionalism.', 'our power will stop all wars and bring a new spirit of unity to a world that has been angry, violent, and totally unpredictable.', 'america will be respected again and admired again, including by people of religion, faith, and good will.', 'we will be prosperous, we will be proud, we will be strong, and we will win like never before.', 'we will not be conquered, we will not be intimidated, we will not be broken, and we will not fail.', 'from this day on, the united states of america will be a free, sovereign, and independent nation.', 'we will stand bravely, we will live proudly, we will dream boldly, and nothing will stand in our way because we are americans.', 'the future is ours, and our golden age has just begun.', 'thank you.', 'god bless america.', 'thank you all.', 'thank you.', 'thank you very much.', 'thank you very much.', 'thank you.']